

A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

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Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W' framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W' . This lumps two physiologically distinct anaerobic systems—the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system—into one tank with a single recovery time constant. That simplification discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will return. This paper develops a **dual-tank model** that splits W' into a fast reserve and a slow reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and ^{31}P -MRS literature. The model is explicitly a physiologically-motivated decomposition of one measured quantity (W^{bal}) whose split is assumed rather than measured, and whose extra resolution over single-tank W^{bal} is concentrated in the transients after hard efforts. It generalises W^{bal} in depletion and diverges from it, by design, in recovery. The model is reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and a Garmin Connect IQ implementation displays live reserve, consumption, and depletion for both systems. We give a validation strategy—including the out-of-sample test that would earn the second tank and a compartment-level test the current model fails—and an explicit statement of assumptions and limits.

Key points.

- The two bars are **compartments of W'** , named for their dominant metabolic system; the metabolic identification is motivation, not a measured claim— W' recovers $\sim 4\times$ faster than the muscle metabolites the tanks are named after (Section 4.2).
- Non-redundant content lives in *transients*: the glycolytic activation ramp at effort onset and the fast-reserve recovery transient after hard efforts. During steady supra-CP effort both bars are affine in W^{bal} .
- Under an oxidative-headroom recovery gate, the effective fast-reserve recovery constant is 100–500 s while a rider is pedalling, so the fast-reserve bar is *live* through the recovery valleys of an interval session.
- For training prescription, the cumulative per-system load separates an alactic from a glycolytic session; the discriminating mechanism is the **glycolytic flux ceiling scaled by the activation ramp**, the metric is blind to *submaximal* alactic work, and it must be judged against a zero-parameter effort-duration baseline, not only W^{bal} (Section 6.2).
- The compartment-level partition is testable in-modality by muscle biopsy during cycle ergometry; against repeated-sprint data the current model fails in level and direction, so the per-system split is a descriptive W' -statistic, not a validated bioenergetic ATP partition (Section 6.3).

Keywords: critical power; W' ; phosphocreatine; anaerobic glycolysis; W^{bal} ; repeated-sprint; real-time cycling analytics; Connect IQ.

1 Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body’s energy-supply systems. The dominant bridge in cycling—the CP/ W' model—treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill.¹ It is power-native, validated, and computationally trivial, which is why it underlies W^{bal} displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: W' is one tank. Physiologically, anaerobic capacity is at least two tanks with very different dynamics:^{2,3}

- The **phosphocreatine (PCr / alactic)** system—high power, low capacity, *fast* oxidative recovery (seconds to a minute), with recovery that is pH-sensitive and slows across repeated bouts.
- The **glycolytic (lactic)** system—larger capacity, lower peak power, and *slow* recovery (minutes) that effectively proceeds only at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the fast reserve is full but the slow reserve is depleted, a 5 s jump is fine but a 40 s dig is not—a distinction single-tank W^{bal} cannot express. The goal is a model that keeps the power-native, on-device virtues of W^{bal} while restoring the two-system resolution.

Two use cases, two outputs. The first—*pacing/racing*—uses the live bars in the moment (“can I cover this attack?”); its value is concentrated in the post-effort transients (Section 7) and is real but narrow. The second—**training prescription and load monitoring**—uses the *cumulative per-system load* over a session, and is where the decomposition is most clearly worth more than W^{bal} (Section 6.2). The two anaerobic systems carry very different physiological and recovery costs: alactic surges—short, high-power, with adequate recovery—resynthesise in seconds to a minute and can be repeated at high volume with little systemic fatigue; glycolytic surges accumulate H^+/P_i and clear over many minutes (lactate $t_{1/2} \approx 1366 \text{ s}$ ⁴) and cost far more recovery. An athlete targeting the PCr system—alactic power, repeat-sprint ability—must keep efforts short and recoveries full enough that glycolysis stays disengaged, and needs to know, per session, how much of the anaerobic work was alactic versus glycolytic. W^{bal} collapses that to one number; the two-tank model does not.

Two honest qualifications. First, the split is a modelling assumption, not a measurement: power alone cannot identify how W' divides between the two systems (Section 4.3). Second, the extra resolution is concentrated in transients. Whenever the fast reserve is full, the slow reserve is merely an affine rescaling of single W^{bal} . The fast reserve is *not* full most of the time while riding, however: under the oxidative-headroom recovery gate (Section 4.3) its effective recovery constant is 100 s to 500 s at typical soft-pedal powers rather than the 27 s of a stopped rider, so the bar stays live through the recovery valleys of an interval session (Section 6.1). The model earns its keep as a heuristic decomposition of W' , not as two independently measured reserves.

2 Physiological background

Phosphocreatine (alactic). ATP is buffered by the creatine-kinase reaction ($\text{PCr} + \text{ADP} \rightarrow \text{ATP} + \text{Cr}$). Intramuscular PCr ($\sim 20\text{--}25 \text{ mmol kg}^{-1}$ wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: the classic alactic O_2 -debt half-time is $\approx 30 \text{ s}$,² and ^{31}P -MRS gives recovery time constants $\tau_{\text{PCr}} \approx 20 \text{ s}$ to

40 s, strongly **pH-dependent** (acidosis slows it) and biphasic under high glycolytic load.^{5,6} Recovery also slows progressively across repeated hard bouts.

Glycolytic (lactic). Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~ 10 s–2 min. Its “fuel level” is best tracked as accumulated muscle/blood lactate, a proxy for H^+ and metabolite accumulation. Critically, glycolytic flux itself *falls* across repeated maximal sprints: acidosis inhibits phosphorylase and phosphofructokinase, so the glycolytic ATP contribution decays from $\approx 40\%$ of total ATP in the first of 10×6 s efforts to $< 10\%$ by the last,^{7,8} while the aerobic contribution rises to absorb the shortfall ($\sim 29\% \rightarrow 43\%$ between sprints).⁹ Restoration of the glycolytic reserve is slow (lactic O_2 -debt half-time ≈ 15 min) and proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1). The model as formulated has no term for glycolytic flux falling with fatigue; this is examined directly, with an optional corrective term, in [Section 6.3](#).

The onset that matters. Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly, while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated,¹⁰ so the early demand is PCr-weighted and shifts toward a shared draw as glycolysis engages.¹¹ This graded onset—and the very different refill rates—is what the dual-tank model must reproduce (via a parallel, PCr-weighted draw and a glycolytic activation ramp; [Section 4.3](#)), rather than a strict “PCr first, then glycolytic” hand-off.

What the two “tanks” represent. The tank is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity. The fast reserve corresponds to a real, depletable substrate store (intramuscular phosphocreatine), so “empty” is close to literal. The slow reserve is better understood as *tolerance to accumulating fatigue-related metabolites* than as a fuel gauge: on the 1–4 min timescale it represents, muscle glycogen is nowhere near exhausted (carbohydrate depletion is a separate, hours-long limiter this model does not address). What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (P_i), H^+ , ADP, and extracellular K^+ toward the limit of tolerance, consistent with exhaustion at CP/W' coinciding with a reproducible low-PCr / high-metabolite state (lactate itself is largely a fuel and a marker, not the cause of force loss). This is why the two recovery laws differ: the fast reserve resynthesises quickly, whereas the slow side recovers slowly and only at low intensity because it reflects clearance and buffering of accumulated byproducts, which need aerobic metabolism and time. In short: **the fast reserve is a fuel that depletes; the slow reserve is a byproduct bucket that fills.**

3 Prior models

Three model families exist:

- **Family A — CP/ W' -balance.**^{1,12,13} Power-native and on-device-friendly, but one lumped tank with a single recovery τ . The recovery law $\tau_{W'} = 546 e^{-0.01 D_{CP}} + 316$ s cannot represent fast PCr and slow glycolytic recovery simultaneously.
- **Family B — hydraulic tank models.**^{14,15} Two explicit anaerobic vessels (phosphagen, glycolytic) plus an aerobic vessel; out-predicts W'_{bal} for intermittent recovery. Its eight parameters are abstract and fit per athlete by evolutionary computation—heavy for an embedded target.

- **Family C — bioenergetic supply/demand ODEs.**¹⁶ Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

The gap. Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about “lighter”: the model’s core count ($CP, W', f_p, P_{p,\max}, g_{\text{rate}}, \tau_p, \tau_g, \tau_{\text{on}}, \text{LT1}$, with $\tau_{\text{off}} = \tau_{\text{on}}$ by default—nine free numeric parameters, plus three optional realism terms and two functional-form gates, gate_p and the intensity-dependent glycolytic recovery shape) is essentially the hydraulic model’s eight: the same size, not a smaller model. The difference is *where the parameters come from*. Only CP and W' are fitted per athlete (from a standard CP test); the rest are hard-coded to literature defaults rather than fit by evolutionary computation. That buys on-device simplicity and a trivial calibration, at a cost: the load-bearing defaults ($f_p, P_{p,\max}, \text{LT1}$, and the recovery τ ’s) are exactly the quantities a CP test does *not* determine, so the two-system resolution rests on assumptions, not measurements. This is a legitimate identifiability-for-convenience trade, and the sections below are explicit about which numbers are earned and which are assumed.

4 The dual-tank model

4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity: the **fast reserve** R_p (capacity C_p) and the **slow reserve** R_g (capacity C_g), with total anaerobic capacity $W' = C_p + C_g$. We call them the fast reserve (“PCr”) and slow reserve (“glycolytic”); the metabolic labels are motivation, not a measured claim. As [Section 4.2](#) shows, the reserves are compartments of W' , with the correct internal fast/slow ratio ($\sim 18\times$) but an absolute timescale $\sim 4\times$ faster than the muscle metabolites they are named after, because W' itself recovers $\sim 4\times$ faster than PCr or lactate. The bar is therefore a fast-recovering component of W'_{bal} , and the model is right about W' ’s internal dynamics while remaining agnostic about the metabolites.

[Table 1](#) lists the parameters. Derived quantities: $C_p = f_p W'$, $C_g = (1 - f_p) W'$, with initial state $R_p = C_p$, $R_g = C_g$ (full). Nine parameters are free numeric values; two functional-form gates (gate_p and the glycolytic recovery shape) and three optional realism terms (pH-slowed PCr recovery, aerobic ramp, glycolytic flux fatigue) sit on top. Only CP and W' are fitted per athlete. A fast-reserve depletion kinetic, $\tau_{\text{dep}} = C_p / P_{p,\max}$, also follows from the table ([Section 4.5](#)) and is equally assumed.

On f_p . We set **0.25** from physiological alactic-fraction estimates (the PCr system supplies ~ 20 – 30% of anaerobic ATP capacity^{3,17}); once the reconstitution reference protocol’s recovery power is known (20 W, near-passive), the reconstitution curve independently implies $f_p \approx 0.20$ – 0.25 . Treat it as a per-athlete calibration target on the band ~ 0.20 – 0.25 .

On $P_{p,\max}$. $P_{p,\max} \approx P_{1s} - CP$ is the rate at a *full* tank. Fast-reserve flux is not constant: the rate *ceiling* tapers with tank fullness (creatine-kinase equilibrium—flux falls as the store depletes), so the available rate is $P_{p,\max} (R_p / C_p)$, equal to $P_{p,\max}$ only when the tank is full ([Section 4.3](#)). Read $P_{1s} - CP$ as an upper bound: at 1 s glycolysis is already $\sim 15\%$ active, so it mildly over-attributes to the fast reserve.

Table 1. Model parameters. Only CP and W' are fitted per athlete; the rest are literature-set.

Symbol	Meaning	Typical default
CP	critical power (W)	from a 3-/12-min CP test ($\approx$ FTP + a few %)
W'	total work above CP (J)	from the same test (~ 15 – 25 kJ)
f_p	fraction of W' in the fast reserve	0.25 — assumed (physiology; Section 7)
$P_{p,\max}$	fast-reserve peak power above CP (W), at a <i>full</i> tank	$\approx P_{1s} - \text{CP}$ (best 1 s power – CP)
g_{rate}	glycolytic peak flux as a fraction of fast-reserve peak flux	0.5 — load-bearing ; ³ band ~ 0.3 – 0.7 , drives the training-load separation (Section 6.2)
τ_p	fast-reserve recovery time constant (s); a W' -recovery constant, not muscle-PCr resynthesis	27 — set by the W' -reconstitution curve (Section 4.2); do not read as biopsy PCr
τ_g	slow-reserve recovery time constant (s)	470 — reconstitution fit ^{4,18} (Section 4.2)
τ_{on}	glycolytic activation time constant (s)	6 ¹⁰
τ_{off}	glycolytic de-activation time constant (s)	$= \tau_{\text{on}}$ — assumed; not load-bearing (Section 6.2)
LT1	first lactate threshold power (W); sets the band of the recovery shape	<i>measured</i> (a threshold test), not a fixed %CP
gate_p	oxidative-headroom recovery gate ($(\text{CP} - P)/\text{CP}$ — a functional form, no free constant)	assumed; shape is a choice (Section 7)
g_{fat}	<i>optional</i> glycolytic flux-fatigue exponent, $\text{rate}_g (R_g/C_g)^{g_{\text{fat}}}$	0 (off) by default (Section 6.3)

On g_{rate} . The ratio $g_{\text{p,max}} = 0.5 P_{\text{p,max}}$ is not arbitrary: peak glycolytic ATP flux is roughly half peak PCr/creatine-kinase flux in human muscle,^{3,10} so the fast system is higher-power by about 2:1. g_{rate} is nonetheless load-bearing: the ablation in Section 6.2 shows it (via the glycolytic flux ceiling) generates the training-load separation, and sweeping it 0.3→0.7 moves that separation from ~48 to ~33 points. It is banded (~0.3–0.7) and should be checked against sprint biopsy data (Section 6.3), against which it has never been fit.

4.2 Reconstitution: calibration source and a tension it raises

The 37 / 65 / 86% W' -recovery at 2 / 6 / 15 min (half-time 234 s) is from Ferguson et al.,⁴ reviewed by Chorley & Lamb,¹⁸ who state it was measured when recovering at a nominal 20 W. Three things follow.

(1) **τ_g is calibrated to ~470 s.** The recovery gate makes the effective constant depend on recovery power; the 20 W figure resolves this. But 20 W is not quite gate = 1: with $\text{LT1} \approx 0.8 \text{ CP} \approx 200 \text{ W}$, gate = $(200 - 20)/200 = 0.90$, so a model reproducing Ferguson needs $\tau_g \approx 470 \text{ s}$.

(2) **f_p is consistent with the curve, not independently measured by it.** With recovery power fixed, the curve constrains a ridge in (f_p, τ_g) , not f_p alone: both $(f_p 0.25, \tau_g 520)$ and the joint optimum $(f_p 0.20, \tau_g 470)$ fit 37/65/86 to within ~3–5 points. Physiology selects $f_p \approx 0.25$ on that ridge; the recovery data alone would prefer ~0.20.

(3) **The tanks are compartments of W' , not of muscle metabolites.** Ferguson reports the two recovery channels underlying W' reconstitution: VO_2 $t_{1/2} = 74 \text{ s}$ (a proxy for oxidative/PCr resynthesis) and blood-lactate $t_{1/2} = 1366 \text{ s}$. Placing these beside the model's component half-times (Table 2) shows the model's two constants are a near-uniform ~4× rescaling of Ferguson's channels—the *shape* (the 18-fold fast/slow separation) is reproduced almost exactly, only the *scale* differs, and by the same factor in both. That is not the signature of fitting two exponentials to a sum. The offset is unavoidable: building the two-compartment model from Ferguson's *measured* channel constants (107 s, 1971 s) cannot reproduce Ferguson's own recovery curve (Table 3)—a metabolite-calibrated model predicts half the observed recovery. W' simply recovers ~4× faster than the PCr/lactate it correlates with, consistent with Ferguson's conclusion that W' reconstitution is “not a unique function of phosphocreatine concentration or arterial [lactate]... unlikely to simply reflect a finite energy store.” What differs from the metabolites is therefore the *naming*, not the time constants. Symmetrically, blood lactate $t_{1/2}$ is no more a measure of *muscle* H^+/P_i clearance than VO_2 off-kinetics is a measure of PCr resynthesis, so neither channel measures a tank and neither adjudicates the decomposition. The fast/slow-reserve naming is the resolution.

Table 2. The model's recovery components against Ferguson's measured channels: the ratio is preserved, the absolute scale is ~4× faster.

Component	Model	Ferguson channel	Ratio
fast	$\tau_p = 27 \text{ s} \rightarrow t_{1/2} = 18.7 \text{ s}$	VO_2 74 s	4.0×
slow	$\tau_g = 470 \text{ s} \rightarrow t_{1/2} = 326 \text{ s}$	lactate 1366 s	4.2×
ratio (slow/fast)	17.4×	18.5×	within 6%

This reconstitution tension does not reach the training-load result of Section 6.2: that signal is driven by the *depletion* law, anchored in measured activation kinetics (τ_{on} from phosphorylase activation¹⁰) and the flux ratio (g_{rate} ³), not by the recovery constants. Substituting Ferguson's own $\tau_p = 107 \text{ s}$ moves the alactic/glycolytic separation by <1 point.

Table 3. A model built from Ferguson’s own channel constants under-predicts Ferguson’s own recovery curve by roughly half; the calibrated model reproduces it.

	2 min	6 min	15 min
Observed ⁴	37%	65%	86%
Model (τ_p 27, τ_g 470)	40%	63%	87%
Built from Ferguson’s channels (107 s, 1971 s)	20%	35%	50%

4.3 The 1 Hz update

Let P be instantaneous power (W) and $\Delta t = 1$ s; define demand relative to aerobic supply as $\Delta = P - CP$.

Case 1 — depletion ($\Delta > 0$, above CP). Demand need = $\Delta \Delta t$ (J) is met by both systems in parallel. Glycolytic supply is not instantaneous: glycogen phosphorylase transforms to its active form over the first several seconds of maximal effort (rising from 12% at rest to 47% by 6 s¹⁰), while phosphocreatine is the immediate buffer.^{9,19} Both contribute from onset, not in sequence.¹¹ A glycolytic activation term $g \in [0, 1]$ ramps first-order with time constant $\tau_{on} \approx 6$ s above CP and relaxes during recovery. The share of submaximal demand is apportioned by *capacity*; a separate peak-flux *ceiling*, with the fast-reserve rate tapered by fullness, governs maximal efforts:

```

g      <- g + (1 - g)*(1 - exp(-dt/tau_on))    # glycolytic activation ("inertia")

# --- SHARE of submaximal demand: capacity-proportional (not rate-proportional) ---
w_p    = C_p                                  # g=0 -> fast covers all; g=1 ->
      fast covers f_p
w_g    = C_g * g
share_p = need * w_p / (w_p + w_g)             # guard: if w_p+w_g ~ 0, share_p =
      need
share_g = need - share_p

# --- RATE CEILING: peak flux, fast-reserve tapered with fullness (maximal efforts
      only) ---
rate_p  = P_p_max * (R_p / C_p)                # guard: C_p > 0
rate_g  = g_rate * P_p_max * g

take_p  = min(share_p, R_p, rate_p*dt)          # each reserve capacity- AND rate-
      limited
take_g  = min(share_g, R_g, rate_g*dt)
unmet   = need - take_p - take_g
take_g += min(unmet, R_g - take_g, rate_g*dt - take_g); unmet -= ... # spill to
      partner,
take_p += min(unmet, R_p - take_p, rate_p*dt - take_p); unmet -= ... # then to a
      deficit
R_p    -= take_p; R_g -= take_g
D      += unmet                                # DEFICIT: supra-CP work the caps
      could not place
if unmet > 0: rate_limited = true                # power beyond the reserves' flux (
      often a stale P_1s)

```

Why the share and the ceiling are different objects. The ceiling ($P_{p,max}$, $g_{p,max}$) encodes “the fast system is higher-power,” true at maximal effort where both systems are flux-limited. It is the wrong object for apportioning *submaximal* supra-CP demand, where neither system is near

its flux limit and sharing is set by metabolic control. Weighting the share by capacity while keeping the taper on the ceiling gives the correct behaviour in both regimes:

- **Submaximal:** both reserves drain $\approx$ proportionally to capacity, so $R_p/C_p \approx R_g/C_g \approx W^{\text{bal}}$ and both reach their nadir together at exhaustion. During a steady supra-CP effort the two bars track W^{bal} and each other; they diverge only in the activation ramp and in recovery.
- **Maximal:** the tapered ceiling makes R_p decay $\sim$ geometrically to its nadir (a ~ 10 s all-out sprint leaves the fast reserve at $\approx 20\text{--}30\%$; [Section 4.5](#)), and fast-reserve dominance emerges from the ceiling, not the weight. This residual is a fast-reserve-of- W' level, motivated by—but not validated against—muscle-PCr biopsy values.
- **Energy conserved when a cap binds.** Residual unmet is banked in a deficit D , so $(R_p + R_g - D)$ drops by exactly $\Delta \Delta t$ per second. Work booked to D never drains a reserve, so a large D leaves the two bars reading slightly full; D is an accounting term for the rare supra-flux case, usually signalling a stale P_{1s} .

The depletion split is not identifiable from power: total draw above CP is all the pedals see, and any $(f_p, g_{p,\text{max}})$ summing to the same total is observationally equivalent. In principle the bi-exponential recovery of W' constrains the split; in practice that fit is ill-conditioned (the fast component is nearly invisible at standard reconstitution sampling), so f_p is assumed and weakly constrained, not recovered.

Case 2 — restoration ($\Delta \leq 0$, at/below CP). Each reserve refills exponentially; the fast reserve is gated by oxidative headroom, and the slow reserve and deficit recover at an intensity-dependent rate:

```
g <- g * exp(-dt/tau_off)           # glycolytic de-activation (tau_off
  = tau_on default)

# FAST RESERVE: resynthesis needs aerobic ATP ABOVE the ride's own demand, so it is
  gated by the
# OXIDATIVE HEADROOM (CP - P): full at rest, near-arrested at CP. gate_p scales the
  AMOUNT (not exponent).
gate_p = max(0, (CP - P) / CP)
R_p += gate_p * (C_p - R_p) * (1 - exp(-dt/tau_p))

# SLOW RESERVE and deficit recover whenever P < CP at the intensity-dependent W'bal
  rate
# tau_W'(CP - P) = 546*exp(-0.01*(CP - P)) + 316, amplitude re-anchored so the 20 W
  passive
# rate reproduces the reconstitution curve. Bounded, and fitted across recovery
  powers.
if P < CP:
  f = gate20 * tau_W'(CP - 20) / tau_W'(CP - P) # = gate20 at 20 W anchor, <
    1 above
  b = f * (1 - exp(-dt/tau_g))
  R_g += (C_g - R_g) * b
  D -= D * b
```

The intensity-dependent recovery shape. A linear gate $(LT1 - P)/LT1$ that falls to zero at $LT1$ makes the effective recovery constant diverge as $P \rightarrow LT1$; under such a gate the model cannot complete a standard $4 \times [4 \text{ min}@330 \text{ W}/4 \text{ min}@150 \text{ W}]$ session (see [Section 4.5](#)). Adopting the established W' -recovery form $\tau_{W'}(D_{CP}) = 546 e^{-0.01(CP-P)} + 316$ ¹—already fitted across recovery powers—for the shape, with its amplitude re-anchored so the 20 W passive rate reproduces the

reconstitution curve, yields bounded, monotone recovery that tracks the reference law within $\sim 15\%$ everywhere and completes the session (Table 5).

The oxidative-headroom gate. $\text{gate}_p = (\text{CP} - P)/\text{CP}$ prevents the fast-reserve bar refilling while the rider is under load. Its shape is a modelling choice: a linear, $\sqrt{\cdot}$, or sigmoid form swings the 60 s post-effort recovery number by 21–31 points at LT1 (46 / 65 / 34% after a 700 W/20 s effort). It also scales the refill only and adds no sub-CP *drain*, so a rider holding 0.95 CP for 30 min—never above CP—keeps a full fast-reserve bar; the structurally correct remedy is to relax R_p toward an intensity-dependent steady state $R_p^*(P)$, which is stated as a limitation (Section 7). After a 20 s/700 W effort (fast reserve $\rightarrow 23\%$), 60 s of recovery restores it to 87% at 0 W, 46% at 0.8 CP, and 25% at 0.99 CP.

The g de-activation lets the activation ramp re-fire on each interval of a repeated-bout set (otherwise g ratchets to 1 on the first surge). $\tau_{\text{off}} = \tau_{\text{on}}$ is assumed: activation, not de-activation, was measured.¹⁰ LT1 sets the anchor band of the recovery shape rather than a hard on/off wall; recovery proceeds for any $P < \text{CP}$, fastest at low intensity and slowing smoothly through the tempo band. It is a measured threshold in watts; where only CP is known, $\text{LT1} \approx 0.80 \text{ CP}$ is a fallback and the weakest default in the model.

Optional realism (off by default).

- *pH-slowed PCr recovery:* $\tau_{p,\text{eff}} = \tau_p (1 + k(1 - R_g/C_g))$. It cannot be pushed to reproduce muscle-PCr recovery ($\sim 78.7\%$ at 3.8 min⁹) without breaking the reconstitution curve and the 4×4 (the $k \approx 16$ that hits the biopsy value drops the curve to 30/68/94 and the 4×4 to -30%)—the same point as Section 4.2: W' and muscle PCr do not share a time constant.
- *Aerobic ramp* ($\tau_{\text{aer}} \approx 25$ s): a first-order aerobic supply $A(t)$ rising toward $\min(P, \text{CP})$, with $\text{need} = (P - A) \Delta t$, reproducing the onset O_2 deficit; it also absorbs the shortfall across repeated sprints (Section 6.3).
- *Glycolytic flux fatigue* (g_{fat}): scale the glycolytic ceiling down as the slow reserve empties, $\text{rate}_{g,\text{eff}} = \text{rate}_g (R_g/C_g)^{g_{\text{fat}}}$, capturing acidotic inhibition of phosphorylase/PFK (Section 6.3).

4.4 Reported quantities

- **Fast-reserve level** $= R_p/C_p$ (0–100%) — “punch” remaining.
- **Slow-reserve level** $= R_g/C_g$ (0–100%) — “sustained dig” remaining.
- **Consumption rate** (per system, W) — $\text{take}_p/\Delta t$, $\text{take}_g/\Delta t$.
- **Combined W^{bal}** $= (R_p + R_g - D)/W'$ — energy-conserving in depletion (matches a single-tank W^{bal} reference on above-CP segments) and deliberately divergent in recovery.
- **Two distinct flags:** `rate_limited` (power beyond the reserves’ flux caps, $\text{unmet} > 0$, reserves not empty—usually a stale P_{1s} , triggering a “recalibrate sprint power” hint) and `exhausted` (both reserves at/near zero). These are different conditions: a fresh rider with a mis-set P_{1s} can be rate-limited but not exhausted.

4.5 Model behaviour

A short, very hard surge draws mostly from R_p and refills afterward at a rate set by recovery intensity. A maximal ~ 10 s sprint leaves $R_p \approx 20\text{--}30\%$ (not empty) from the emergent depletion constant $\tau_{\text{dep}} = C_p/P_{p,\text{max}}$; because τ_{dep} swings $\sim 4\text{--}13$ s across riders, the 10 s residual ranges from $\sim 10\%$ to $\sim 46\%$. A sustained supra-CP effort draws from both reserves, declining together to their nadir; the drawdown is front-loaded, not linear (450 W: 68/42/18/1 at quarter-TTE

intervals). During such steady efforts both bars are affine in W^{bal} , so their only non-redundant content is the ramp offset and the post-effort recovery transient.

For a constant supra-CP effort, R_p/C_p is asymptotically a function of the fraction φ of W' spent, but only for efforts long relative to τ_{on} and below the ceiling regime; the activation ramp adds a second intensity dependence, so “hold different supra-CP powers to exhaustion and compare fast-reserve-at- φ ” is a *falsification* test (a pass means the bar is a deterministic restatement of W^{bal} in depletion), not a validation.

Table 4 reports a synthetic battery at CP = 255 W, $W' = 20$ kJ, $f_p = 0.25$, $P_{p,\text{max}} = 690$ W ($g_{\text{rate}} = 0.5$), $\tau_p = 27$, $\tau_g = 470$, $\tau_{\text{on}} = \tau_{\text{off}} = 6$, LT1 = 204 W. Table 5 shows the recovery law’s behaviour across recovery powers: the intensity-dependent shape is bounded and tracks the reference W' -recovery law within $\sim 15\%$, and the $4 \times [4 \text{ min}@330/4 \text{ min}@150]$ session completes (W^{bal} at rep starts 100/49/7/ – 24%).

Table 4. Synthetic verification battery.

Test	Result
Sustained 450 W — fast reserve at 25/50/75/100% of TTE	68 / 42 / 18 / $\sim 1\%$ (front-loaded)
Both reserves at exhaustion	fast $\sim 1\%$ / slow $\sim 0\%$ (empty together)
945 W (= P_{1s}) 10 s sprint — fast-reserve residual	23% (ceiling-dominated)
$6 \times [10 \text{ s}@700/30 \text{ s}@150]$ — fast reserve per bout	$49 \rightarrow 17 \rightarrow 10 \rightarrow 8 \rightarrow 8 \rightarrow 7\%$
1200 W hold (caps bind) — W^{bal} conservation	leak = 0 J
Fast-reserve recovery after 700 W/20 s, 60 s @ {0,128,204,242,252} W	87 / 68 / 46 / 30 / 25%
Debt repayment at 0.9 CP	D clears slowly (recovery-law rate), not frozen

Table 5. W^{bal} recovery half-time after full depletion, holding a constant recovery power: the intensity-dependent shape tracks the reference law and stays bounded.

Recovery power	Linear-gate half-time	Intensity-dependent shape	Reference $\tau_{W'}$
20 W (anchor)	201 s	201 s	255 s
100 W (39% CP)	353 s	241 s	299 s
150 W (59% CP)	675 s	291 s	351 s
190 W (75% CP)	2601 s	367 s	417 s
204 W (= LT1)	∞	409 s	446 s

In depletion the model is a faithful generalisation of W^{bal} ($(R_p + R_g - D)$ falls by exactly $\Delta \Delta t$ per second); in recovery it is deliberately different (bi-exponential, intensity-gated), so it does not reduce to a single $\tau_{W'}$. It is a generalisation in structure and depletion behaviour, not a strict superset.

5 Real-time on-device implementation

The per-second cost is a handful of multiplies and one $\exp()$ per reserve—trivial for a head unit. The target design is a Garmin Connect IQ full-screen *Data Field* (`Toybox.WatchUi.DataField`), called once per second (a natural fit for $\Delta t = 1$ s), reading `Activity.Info.currentPower`. Configuration (CP, W' , f_p , $P_{p,\text{max}}$, LT1, and the recovery constants) is exposed via Connect IQ application properties. The display is two vertical bar gauges (fast and slow reserve) with numeric % and colour bands; per-system live consumption is a *modelled share*, not a reading,

because when the ceiling is slack it is merely $(P - \text{CP})$ split by a fixed fraction (Section 7), and is labelled as such or omitted. Both reserves and consumption are written to the FIT file via `FitContributor` fields for post-ride analysis. The state is a handful of scalars (R_p , R_g , the activation g , the deficit D)—no arrays—well within Connect IQ memory budgets.

6 Validation strategy

The model is a hypothesis and must be checked against data. Its headline feature—a separate fast reserve tracked in real cycling—is the hardest thing to validate, so we separate tests that check the plumbing from the one test that would earn the second tank.

1. **Face validity.** Synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours of Section 4.5.
2. **Backward compatibility (depletion only).** The combined balance ($R_p + R_g - D$) reproduces a reference W^{bal} implementation in depletion (it drops by exactly $\Delta \Delta t$ per second, deficit-preserved even when a cap binds). It will not match in recovery, by construction, so the test is scoped to above-CP segments.
3. **Reconstitution targets.** Combined recovery after a full W' depletion is compared to the published curve (37/65/86% at 2/6/15 min⁴). The curve is blind to the fast reserve (a $\tau_p \approx 27$ s process is $\sim 99\%$ complete by the first 2 min sample), so it validates slow-reserve recovery and the size of the fast offset, not fast kinetics; and it fixes τ_g while constraining a (f_p, τ_g) ridge, not f_p alone (Section 4.2).
4. **Fast-recovery kinetics.** The load-bearing test for τ_p is a repeated-bout protocol with early recovery sampling (all-out efforts separated by 10/20/30/45/60 s), with between-bout power recovery correlated with modelled fast-reserve recovery.⁹ Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth, and its limit.** Blood-lactate kinetics can corroborate the slow reserve, but the fast reserve's gold standard, ³¹P-MRS, is essentially unavailable in real cycling (the muscle must be inside a magnet), so it exists only for small-muscle knee-extension or immediate post-exercise. The headline novelty cannot be checked against ³¹P-MRS in the modality it targets.
6. **The head-to-head that earns the second tank.** Because the model is a near-superset of single-tank in depletion, “it fits better” is guaranteed and is not evidence the compartments are real. The decisive test is a head-to-head against single-tank W^{bal} on an out-of-sample intermittent-tolerance prediction.¹⁵ Since depletion here is single-tank by construction, the only lever is the recovery law, so this is a test of the recovery law, not of the two-compartment hypothesis; the protocol should choose recovery-valley powers straddling LT1, where the two recovery laws diverge most.
7. **Field calibration.** Fit $P_{p,\text{max}}$ and the τ 's per athlete to maximal and repeated-bout tests. f_p is not a routine target (the split is unidentifiable from power); any single-tank comparator in the head-to-head must have its own $\tau_{W'}$ fit to the same data, or the dual-tank wins trivially.

6.1 Empirical usefulness on real interval rides

Two data-in-hand questions decide whether the feature is worth building. Both were evaluated on six real interval / VO_2max sessions (the model's best case; a steady endurance ride would score far lower).

Is the fast-reserve bar informative, and is it a function of W^{bal} ? The bar is below 95% (live) for a per-ride 20 / 31 / 38 / 52 / 61 / 79% of ride time (median $\sim 45\%$). Binning W^{bal} and measuring the spread of R_p/C_p within each 1-point bin gives a mean spread of ~ 58 points (max ~ 93); regressing the bar on its best affine W^{bal} predictor gives $R^2 \approx 0.03$, with the bar >5 points off that affine line $\sim 93\%$ of the time. The bar is emphatically not a function of W^{bal} .

Does the depletion machinery change the display, versus a recovery-only null? Running the full model and the null on the same traces, the fast-reserve bar diverges by up to 25–60 points, with maxima at the starts of recovery valleys. Divergence is not accuracy, however: two models differing by 25–60 points tells us the caps *change* the display, not that the changed display is *right*. These results establish that the bar carries information; whether that information is correct is [Item 6.](#) / [Section 6.3.](#)

6.2 Training-load partitioning

The cumulative per-system load over a session—how many kJ came from the fast versus the slow system—is a different functional of the power trace than W^{bal} , and is the strongest reason to build the second tank. For two fully-specified archetypes (alactic: $10 \times [6\text{s}@900/300\text{s}@150]$; glycolytic: $5 \times [60\text{s}@360/120\text{s}@150]$), reporting the deficit D as a third quantity ([Table 6](#)), the model separates the sessions by ~ 39 points. Three qualifications are essential.

Table 6. Per-system load for two session archetypes (default parameters), with the deficit reported.

Session	fast (kJ)	slow (kJ)	D (kJ)	fast % of (fast+slow)
$10 \times [6\text{s}@900]$ — alactic	29.2	8.7	1.2	77%
$5 \times [60\text{s}@360]$ — glycolytic	13.2	21.5	1.8	38%

(a) The mechanism is the glycolytic flux ceiling scaled by the ramp, not the ramp alone. Ablating the model on the alactic session ([Table 7](#)): removing the ceilings costs ~ 30 points, removing the ramp ~ 25 ; neither alone reaches the full 77%. It is an interaction—during a 6 s sprint the glycolytic ceiling caps glycolysis at a few tens of watts while ~ 645 W is demanded, and the spill goes to the fast reserve—so the load-bearing parameter is g_{rate} . Correspondingly, a $20\times$ change in τ_{off} moves the split by only 1–2 points, whereas $g_{\text{rate}} 0.3 \rightarrow 0.7$ moves the separation $48 \rightarrow 33$ points.

Table 7. Ablation on the alactic session: the ceiling and the ramp both contribute; neither alone reaches the full separation.

Configuration	alactic fast %	separation vs glycolytic
full model	77%	~ 39 pts
no activation ramp ($g \equiv 1$)	52%	~ 15 pts
no rate ceilings (share rule only)	47%	~ 9 pts
neither	25%	-12 pts

(b) The metric detects supra-flux-ceiling power, so it is blind to submaximal alactic work. Holding the 10×6 s structure fixed and sweeping power, the reading climbs $52 \rightarrow 62 \rightarrow 66 \rightarrow 72 \rightarrow 77\%$ at 450/550/600/700/900 W. A textbook alactic session at 450 W reads $\sim 52\%$, close to the glycolytic session’s 38%.

(c) The honest baseline is a stopwatch, not W^{bal} . A coach with the power file and no model can compute the fraction of supra-CP work done in efforts under 15 s—100% versus 0% for these two sessions, perfect separation with zero parameters. The open question is therefore

whether the per-system load carries information beyond effort duration and intensity (regressing fast-reserve % on a duration/intensity baseline across real rides answers it). What does survive: the separation is robust to every assumption it rests on (including substituting Ferguson’s $\tau_p = 107$ s, <1 pt), and it is a genuinely different functional of the power trace. Absolute kJ scale with f_p , D must be reported (and the split refused when D is a large fraction of total—e.g. a 4 × 4 books $\sim 45\%$ of its anaerobic work to D), and this is a training-design signal, not a validated training outcome.

6.3 A compartment-level test the model fails

Muscle biopsy during cycle ergometry *is* in-modality, compartment-level data, and repeated-sprint studies partition ATP supply by system from vastus-lateralis biopsies during maximal cycle-ergometer sprints.^{7,8} The model’s alactic protocol (10×6 s maximal sprints) is exactly this design, and the biopsy literature reports the glycolytic ATP share falling from $\approx 40\%$ in the first sprint to $<10\%$ in the last, with aerobic metabolism rising to cover the shortfall. The model fails this test in level and direction (Table 8).

Table 8. Repeated-sprint glycolytic share: the model, as formulated, runs at the wrong level and in the wrong direction relative to biopsy data.

	Sprint 1		Sprint 10
Observed ^{7,8} — glycolytic share	$\sim 40\%$	$\searrow$	$<10\%$
Model, default ($g_{\text{fat}} = 0$)	23%	$\nearrow$	46%
Model, $g_{\text{fat}} = 1$ + aerobic ramp	23%	$\rightarrow$	$\sim 34\%$ (flat, then falling)

The model starts glycolysis at about half the observed share and has it *rise* across bouts, because it has the two fatigue mechanisms backwards: the fast-reserve ceiling ($\text{rate}_p = P_{p,\text{max}} R_p / C_p$) fades as the reserve empties, so more demand spills to glycolysis bout over bout, while glycolytic flux has no fatigue term. The optional g_{fat} term plus the aerobic ramp flatten the trend and pull the late bouts down but do not reproduce $40\% \rightarrow <10\%$; reaching 40% in sprint 1 would require a higher g_{rate} that then wrecks the separation of Section 6.2. Two conclusions follow. This is a genuinely falsifiable result: a number wrong in a specific, fixable way, obtained because the partition finally faced compartment-level data. And it fixes the status of the output: the per-system split is a W' -decomposition heuristic and a descriptive statistic of a power file, not a validated bioenergetic ATP partition. Calibrating g_{rate} to the sprint-1 biopsy value and fitting g_{fat} to the decay—using data already in the literature—is the highest-value experiment remaining and the one test that could directly falsify or earn the two-compartment hypothesis.

7 Assumptions and limitations

- **The split f_p is assumed and only consistent with (not measured by) the recovery data.** Power cannot identify the depletion split; the reconstitution curve constrains a (f_p, τ_g) ridge, and physiology selects f_p on it (Section 4.2). Supported band ~ 0.20 – 0.25 ; outputs are sensitive to it.
- **The reserves are a decomposition of W' , not latent metabolic state.** When the fast reserve is full (rest), the slow bar is an affine transform of W^{bal} . Under the recovery gate the effective fast-reserve constant is τ_p / gate_p —27 s only at a standstill, rising to ~ 44 s at 100 W, ~ 66 s at 150 W, ~ 135 s at LT1—so while riding the bar is live for a substantial fraction of interval-ride time; the “affine” caveat applies to stopped or easy recovery, not to riding.

- **gate_p is an assumption with an undefended shape** (linear vs $\sqrt{\cdot}$ vs sigmoid swings the post-effort recovery number 21–31 points at LT1) and adds no sub-CP drain.
- **g_{rate} is load-bearing** (it generates the training-load separation) and has never been fit to sprint biopsy data (Section 6.3).
- **The fast-reserve depletion rate $\tau_{\text{dep}} = C_p/P_{p,\text{max}}$ is assumed**, is the ratio of three parameters, and swings $\sim 3\times$ across plausible riders ($\sim 4\text{--}13$ s); it should be sanity-checked against literature PCr depletion half-times.
- **τ_{off} is assumed and not load-bearing** (a $20\times$ change moves the training-load split 1–2 points). The discriminating content is the glycolytic flux ceiling scaled by the ramp, not τ_{off} .
- **LT1 should be measured, not derived from CP**. The 0.80 CP fallback is the weakest default and materially changes recovery estimates.
- **Small modelling choices are assumptions, not derivations**: the recovery shape (an established prior fitted across recovery powers, but still a choice of form); lumping deficit clearance and slow-reserve refill onto the same τ_g ; and the spill order (tested inert on the sessions that matter).
- **The deficit D corrupts the display at maximal effort** (large D leaves the bars reading slightly full). Its late-effort recurrence is driven by the fullness taper collapsing the flux ceiling below what a deep-but-still-sprinting rider produces—structural, not a settings error; D should be reported and the per-system split refused when D is a large fraction of total.
- **Identifiability degeneracies must be named before fitting**: f_p and $P_{p,\text{max}}$ both set the sprint residual; τ_p and the pH coefficient both set the repeated-bout decay; g_{rate} and the ramp both set the partition level.
- **Per-system live consumption is a modelled share, not a reading**: where the ceiling is slack, $\text{take}_p = \Delta C_p/(C_p + C_g g)$ contains neither reserve and settles to a fixed Δf_p split—($P - \text{CP}$) rescaled.
- **A structural product fork**. The lightweight-model rationale serves the *pacing* use case on a head unit. The training-load metric (Section 6.2) is a post-hoc computation on a power file; off-device, a better-resolved model (hydraulic or bioenergetic) becomes available and can be calibrated to biopsy data. A live two-bar pacing display and a post-ride training metric have different centres of gravity.
- **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of fast-reserve recovery unless the optional term is enabled.
- **A hard CP boundary** omits the aerobic slow component and onset kinetics unless the aerobic ramp is enabled; expect over-attribution to the anaerobic reserves in long low-intensity recovery.
- **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fibre-type effects. It is a training/pacing aid, **not medical or diagnostic**, and parameters drift with fitness, fatigue, heat, and altitude.

8 Conclusion

The single-tank W' model earned its place by being power-native and light enough to run anywhere, but it answers “how much anaerobic work is left?” without answering “in which system?” The dual-tank model offers a physiologically-motivated split at negligible computational

cost: divide W' into a fast, rate-limited, fast-recovering reserve and a slow, large, slowly-recovering reserve; share supra-CP demand by capacity; cap each by its peak flux (the fast reserve tapering with fullness); and refill them with system-specific, intensity-gated laws. It generalises W^{bal} in depletion, diverges from it by design in recovery, runs at 1 Hz on a head unit, and surfaces a two-system decomposition—*punch left* and *dig left*—most informative in the transients where pacing decisions concentrate.

Two framing points are essential to state plainly. The reserves are compartments of W' , not of muscle metabolites: W' recovers $\sim 4\times$ faster than the PCr and lactate the tanks are named after (Section 4.2), which makes the time constants correct for W' while the metabolic labels remain motivation. And the per-system split is falsifiable in-modality: run against repeated-sprint biopsy data it fails in level and direction (Section 6.3), so it is a descriptive W' -statistic, not a validated bioenergetic partition. The model is therefore best located as a lightweight, honest heuristic decomposition of W' —with a live pacing display whose value rests on the fast-reserve recovery transient, and a post-ride training metric that is promising, descriptive, and, for the first time, has a data-in-hand test that can earn or sink it. The remaining moves are experiments, not arguments: the recovery-law head-to-head (Item 6.) and the biopsy calibration (Section 6.3).

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